

Juan David Munoz Bolanos

Junior System Engineer

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Summary

Junior Engineer with research experience handling optomechatronic and medical devices from concept to prototype. Proven track record in translating physical concepts into functional prototypes using V-model systems engineering. Adept at collaborating in interdisciplinary teams to deliver high quality technical documentation for clinical and industrial applications.

Research Experience

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>PhD Researcher in adaptive optics</i>	Innsbruck, Austria
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End to End System Design: Led the hardware to software implementation of multiphoton microscope, integrating laser sources, galvo scanners, and light modulators. Achieved enhancing optical image quality.

Systems Engineering: instrument development following V-model principles, managing requirements and tracking from concept to component testing.

Quality & Testing: Executed Design of Experiments (DoE) and optical simulations in Python. Applied testing and documentation protocols for optomechanical assemblies frameworks.

CAD & Prototyping: Designed custom optomechanical mounts and alignment tools using SolidWorks and FreeCAD, ensuring precise mechanical tolerances and seamless integration.

Mar 2020 to Dec 2021	Leibniz Institute of Photonic Technology <i>R&D Research Assistant</i>	Jena, Germany
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Spectrometer Testing: Testing and modifying a dispersive 1064 nm fiber probe Raman spectrometer for non-invasive human tissue characterization, handling optomechanical layout and detector calibration.

Data Analysis & ML: Developed a custom Python package (SpectraMap) to apply supervised machine learning algorithms to 2D hyperspectral Raman images, achieving high accuracy classification of tissue samples.

Technical Documentation: Authored peer reviewed publications detailing optical design parameters and calibration procedures.

Technical Competencies

Hardware	Adaptive optics	Multiphoton/confocal microscopy	Raman spectroscopy			
	Laser manipulation	Camera integration	3D CAD (SolidWorks, FreeCAD)			
Software	Python (scikit-learn, NumPy)	LabVIEW/FPGA	C++	OpenCV	Git	MATLAB

Education

Jan 2022 to Jun 2026	Medical University of Innsbruck <i>PhD in Adaptive Optics</i>	Innsbruck, Austria
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Oct 2019 to Sep 2021	Friedrich Schiller Univ. Jena <i>Master of Science in Photonics</i>	Jena, Germany
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* Dark tags: expertise; light tags: knowledge.

Aug 2012 to Aug
2018

University of Cauca
Bachelor of Physics Engineering

Popayán, Colombia

Publications

Muñoz Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. *ACS Photonics* (2025).

Muñoz Bolaños, J. D. et al. Dispersive 1064 nm fiber probe Raman. *Applied Sciences* (2024).

Sohmen, M., Muñoz Bolaños, J. D. et al. Optofluidic Adaptive Optics in multiphoton microscopy. *Biomedical Optics Express* (2023).

Soft Skills

Systems & Quality Engineering: V-Model, 8D Problem Solving, Design of Experiments.

Adaptability & Independent Working: Self-driven approach to bringing concepts to working systems; comfortable in dynamic, fast-paced environments.

Teamwork & Communication: Cross-functional collaboration across optics, hardware, software, and production teams; structured root cause analysis.

Personal Interests

- Biking/Hiking
- Bachata Dancing
- Acoustic Guitar
- Learning German

Languages

- English (Fluent)
- German (Intermediate B2)
- Spanish (Native)

References

Prof. Dr. Monika Ritsch-Marte
Medical University of Innsbruck
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PD Dr. Christoph Krafft
Leibniz IPHT
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